

Decision trees and random forests

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Decision trees, bagging, and random forests are all supervised machine learning (ML) techniques.¹ We start with explaining decision trees using a classification task but note that regression tasks can also be used with decision trees and work similarly. As in the case of the K-nearest-neighbor method,¹ the observations that fall in the regions defined by the tree splits are averaged (in a regression task) or used for majority voting (in a classification task). The collection of these models is also called classification and regression trees in modern literature.²

In this article, we explain how a tree can be classified on the basis of the practical example introduced by Konstantonis et al³ to predict the extraction/nonextraction treatment on the basis of demographic, cephalometric, and model covariates. A mock example of a decision tree is shown in Figure 1. It pertains to deciding whether to extract a tooth based on the mandibular position (MP) and gender. The MP is a continuous explanatory variable (feature) with values between 70 and 90. Gender is a factor (ie, a categorical variable) with 3 categories: female (F), male (M), or other (X). Note that all models discussed below are presented for illustration purposes and are not derived from the models fitted to the data.

The decision tree works as follows. We start at the top of the tree in Figure 1. If $MP \geq 77$, we move to the right of the tree; otherwise, we move to the left. The places in which the splits are made are called the nodes. After a split at a node, we enter another node or reach a decision. In the case of the top node in Figure 1, for $MP \geq 77$, we get a decision: do not extract the tooth. This is the end of the decision tree; it is called a leaf or a terminal node. For $MP < 77$, we enter another node, in which we split according to gender. The split leads to 2 leaves: one for females with the decision not to extract the tooth, and one for males and other genders with the decision to extract the tooth.

The reader is possibly familiar with presenting such decision trees as graphical decision tools.^{4,5} Decision trees can also be used as ML models. It is possible to train an optimal decision tree using a dataset. During training,

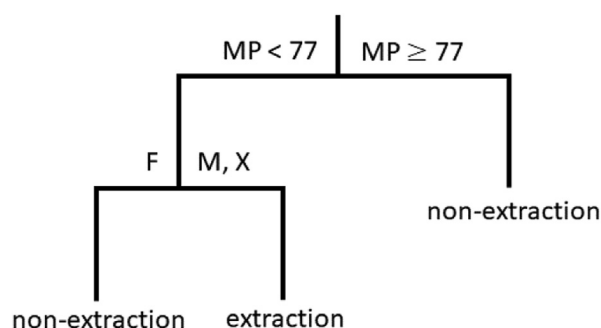


Fig 1. A decision tree that decides whether to extract a tooth. The MP and gender (M, F, and X) are the features in the model.

the decision tree learns which features it should use to perform a split and at what value of the feature it should split. Figure 2, A shows the same decision tree as in Figure 1. In Figure 2, B, we show how we arrived at this decision tree by training it on (fitting it to) training data. Note that the model can create nonlinear decision boundaries.

When training the decision tree, the algorithm applies a split on a feature that results in a region that maximally reduces the classification error. Generally, this does not produce good results. In practice, other measures (such as Gini impurity or entropy) are used to decide the splits. We stop creating new splits when a pre-defined maximum number of splits before the terminal node is reached or when all terminal nodes contain only 1 class.

In previous articles in this series, we discussed overfitting.^{6,7} For decision trees, overfitting usually occurs when the tree is too large with many splits before making a final decision. Overfitting can be managed by forcing the tree to become smaller (ie, by removing nodes near the end of the tree using a process called pruning).

Decision trees possess several interesting properties. They can naturally handle combinations of categorical and continuous features. Our example illustrates this, in which gender is a categorical feature, and MP is a continuous feature. Decision trees are interpretable and often follow human-like intuition when making a decision: the final decision can be reconstructed by

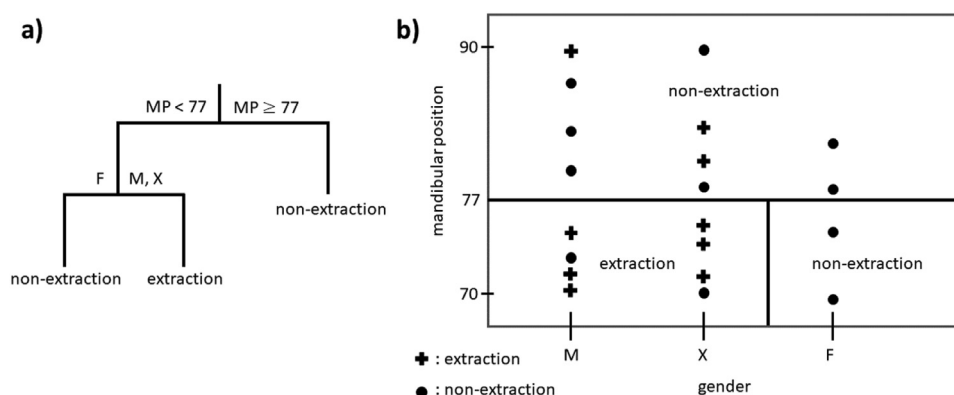


Fig 2. The decision tree (A) is trained on the training data shown in (B). The decision boundary from the decision tree is shown on the plot of the training data. Two rectangular areas predict nonextraction, and one predicts extraction. In this feature space, linear models would result in decision boundaries corresponding to straight lines. The decision boundary is, therefore, nonlinear.

starting at the top of the tree and following the splits. They are robust to outliers and can easily deal with missing data. Despite these advantages, decision trees generally have low predictive performance compared with most other models. Unfortunately, these limitations cannot be compensated by the advantages. However, decision trees form the basis of 2 of the most competitive ML models from a predictive performance perspective: random forests and boosting. We discuss boosting in the next article. In the remainder of the current article, we explain random forests.

Random forests are an example of an ensemble method. Ensembling (ie, averaging the predictions of several models to obtain the final prediction) is a general method beneficial for many ML algorithms, including deep neural networks.⁸

Instead of training a single model in an ensemble, we train several models and average over all the predictions. Individual models can make more mistakes, but as long as the mistakes are different, the average prediction will not be strongly affected, and often, it will be more robust against overfitting. However, we usually only have 1 dataset, and training several times on the same data will lead to the same decision tree. Thus, we have to “force” the decision trees to be different enough so that averaging them makes sense.

The first step to force this difference is to apply the bootstrap procedure discussed in Burzykowski et al.⁷ Decision trees are trained on bootstrap samples obtained from the training data. This set of samples is called a bag. Because of the bootstrapping, the data (ie, the bag) used to train the models are different. Therefore, the resulting decision trees are also different. Next, the results of the different trees are aggregated by averaging the predicted values. The entire procedure of bootstrapping and

aggregating is combined in the portmanteau, bagging. The test error can then be estimated by the out-of-bag error, computed by averaging the prediction errors across all models for observations that were not in the bag. In this case, the out-of-bag error replaces the cross-validation error⁷ to estimate the prediction error.

The second step to force the decision trees to be different is to allow the decision tree to split at each node using a feature chosen from a random subset of the features. A random forest is an ensemble of bagged decision trees forced to make splits on a random subset of the features. The play on words is hopefully clear.

To illustrate the random forest procedure, we extend the list of features used for the task of predicting tooth extraction² using as predictors the variables overbite, overjet, maxillary crowding, and mandibular crowding. At the top of Figure 3, we show the overall decision tree trained on the entire dataset using all 6 features. The tree uses only 3 features: MP, gender, and overbite. It gives a balanced classification accuracy of 67%. A random forest including 100 decision trees yields a higher accuracy of 71%. Figure 3 also includes 2 trees from the 100 random forest ensemble. Their individual classification accuracies are equal to 58% and 64%. Note that for each node, the tree was given a choice of 2 variables, of which one was selected to create a split. The average accuracy across all 100 random forests is higher than that of individual trees.

The predictive performance of a random forest is often close to or equal to that of the best available ML model. Random forests are often a good choice in medium to large datasets, especially given their useful properties regarding missing data and flexibility to include features of mixed types. However, random forests typically do not perform well if the dataset contains

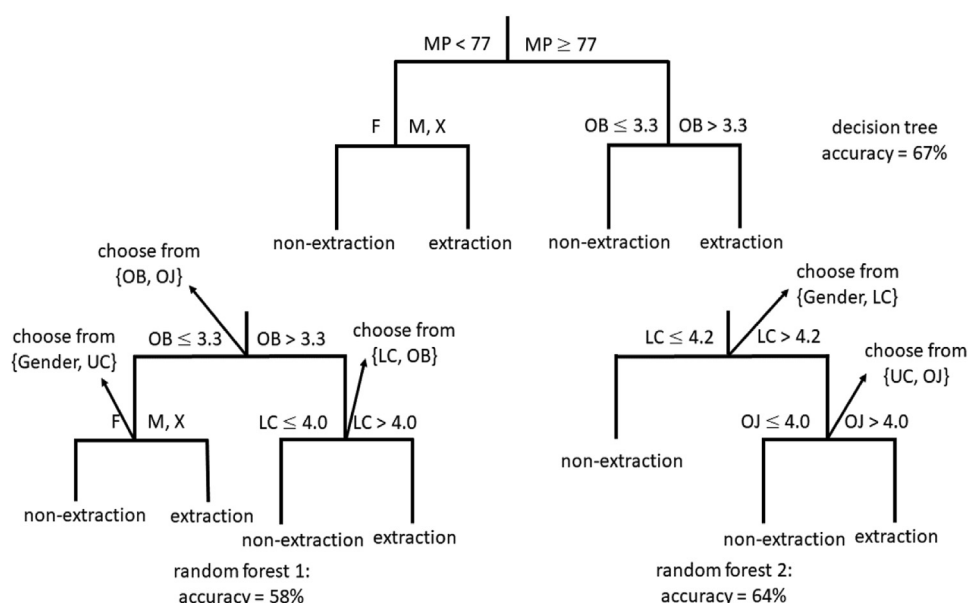


Fig 3. Graphical illustration of the random forest procedure. The estimated accuracies (test errors) are shown for each decision tree: **A**, The decision tree that predicts tooth extraction with access to the features MP, gender, overbite (OB), overjet (OJ), mandibular crowding (LC), and maxillary crowding (UC); **B**, Two decision trees from the random forest ensemble trained on the same data. Because these trees are not allowed to use all features for each split and are trained on a bootstrapped version of the training data, they are different. We use the common rule that the number of features available at each split is the square root of the total number of features. Because we have 6 features in total and $\sqrt{6} = 2.4$, there are 2 available features for each split. The available features are shown at each node.

Table I. Error table for the data of Konstantonis et al,³ for a decision tree model

True Class	Predicted Class	
	Nonextraction	Extraction
Nonextraction	322	74
Extraction	72	73

Table II. Error table for the data of Konstantonis et al,³ for a random forest model

True Class	Predicted Class	
	Nonextraction	Extraction
Nonextraction	372	24
Extraction	70	75

many uninformative features for the response variable. This is because the likelihood of being able to use an informative feature for a split becomes small. In this case, feature selection or dimensionality reduction techniques should be used before training the model.⁹ An important disadvantage of random forests compared with decision trees is difficulty interpreting the results. A typical number of decision trees in a random forest is between 100 and 1000, but it is impossible to have >1000 and fully understand what the model is doing. Some methods can assign an importance score to each feature used in the random forest,¹⁰ which provides insight into how the model works and how to interpret the results.

The performance of a decision tree and a random forest model trained on the data of Konstantonis

et al³ is shown in [Tables I and II](#). The performance of the random forest is superior to that of the decision tree. The evaluation was performed using 5-fold cross-validation.

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